

RFT Web Admin

Backend contract reconstructed from the current admin frontend repository.

Observed

Inferred

Multi-role

Repo-based

What this PDF captures

Every backend requirement below is reconstructed from the real admin UI routes, components, tables, forms, role navigation, mock data, and placeholder actions found in this repository. It separates directly observed behavior from strongly implied backend needs so the build order is grounded in the current web admin surface rather than generic advice.

ROLES

3 workspaces

super admin, school admin, lecturer

FOCUS

Admin operations

onboarding, access, content, admissions

OUTPUT

REST contract

modules, endpoints, workflows, risks

Document Guide

Interpretation rule

Observed items come directly from code. Inferred items are proposed backend contracts that are strongly implied by the UI, flows, and data structures present in the repo.

Front-end Reality Check

- Observed: there is no live API client, no 'fetch', no 'axios', no React Query/SWR layer, no 'app/api' routes, and no backend env configuration in this repo. All business data is mocked in 'lib/mock-data/*' and 'lib/mock-users.ts'.
- Observed: authentication is demo-only and handled with 'localStorage' plus cookies in 'hooks/useAuth.ts' and 'proxy.ts'.
- Inferred: the endpoint paths below are proposed REST contracts chosen to satisfy the observed UI and workflow expectations.

Sections Included

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1. Repo Summary

- Observed: this is a Next.js 16 admin portal with three role-scoped workspaces: 'super_admin', 'school_admin', and 'lecturer' ('types/index.ts', 'lib/navigation.ts', 'hooks/useAuth.ts').
- Observed: the app manages platform-wide institution onboarding, subscription plans, role provisioning, school/faculty operations, lecturer/course oversight, student admissions, content distribution, assessments, and class scheduling ('app/(dashboard)/dashboard/super/*', 'app/(dashboard)/dashboard/school/*', 'app/(dashboard)/dashboard/lecturer/*').
- Observed: the UI is not only a super admin panel. It is a mix of:
 - super admin / multi-institution control
 - school admin / local academic operations
 - lecturer operational workspace
- Inferred: the backend must support multi-tenant RBAC, role-scoped datasets, and shared domain objects that are writable by admins and readable by lecturer/mobile clients.
- Observed: root scripts such as 'generate_backend_pdf.py' and 'generate_student_pdf.py' suggest the product is expected to have a shared backend serving both admin and student/mobile surfaces, but those scripts are documentation generators, not live integrations.

MAIN BUSINESS DOMAINS FOUND IN CODE

- Authentication and session routing: 'components/auth/LoginForm.tsx', 'hooks/useAuth.ts', 'proxy.ts'
- Platform schools/institutions: 'app/(dashboard)/dashboard/super/page.tsx', 'app/(dashboard)/dashboard/super/schools/*', 'components/demo/SchoolActionCenter.tsx'
- Role/access management: 'app/(dashboard)/dashboard/super/roles/page.tsx'
- Subscription plans and billing: 'app/(dashboard)/dashboard/super/fees/page.tsx'
- Faculties/levels: 'app/(dashboard)/dashboard/school/faculties/page.tsx', 'app/(dashboard)/dashboard/school/levels/page.tsx', 'components/dashboard/FacultyTable.tsx'
- Courses and class reps: 'app/(dashboard)/dashboard/school/courses/page.tsx', 'components/lecturer/AssignedCoursesTable.tsx', 'components/demo/ClassRepAssignmentBoard.tsx'
- Lecturer operations: 'app/(dashboard)/dashboard/school/lecturers/page.tsx', 'app/(dashboard)/dashboard/lecturer/*'
- Students and admissions: 'app/(dashboard)/dashboard/school/students/page.tsx', 'app/(dashboard)/dashboard/school/admissions/page.tsx', 'components/demo/AdmissionQueuePanel.tsx'
- Content library and uploads: 'app/(dashboard)/dashboard/*/content/page.tsx', 'components/demo/ContentLibraryPanel.tsx', 'components/dashboard/ContentUploadPanel.tsx'

- Assessments: 'app/(dashboard)/dashboard/lecturer/assessments/page.tsx', 'components/demo/AssessmentBuilderPanel.tsx'
- Class scheduling: 'app/(dashboard)/dashboard/lecturer/schedule/page.tsx', 'components/demo/ScheduleClassForm.tsx'
- Analytics/reporting: 'app/(dashboard)/dashboard/super/analytics/page.tsx', 'app/(dashboard)/dashboard/lecturer/performance/page.tsx'
- Support/settings/logs placeholders: 'app/(dashboard)/dashboard/lecturer/help/page.tsx', 'app/(dashboard)/dashboard/lecturer/settings/page.tsx', 'components/layout/Sidebar.tsx', 'components/layout/MobileSidebar.tsx'

2. Module Inventory

Module	Scope	Access	Evidence	Notes
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Authentication & Session	Admin login, session restore, logout, route gating	Public + all roles	'components/auth/LoginForm.tsx', 'hooks/useAuth.ts', 'proxy.ts'	Currently mocked only
Platform Schools	Institution directory, onboarding, profile, deactivation, export	'super_admin'	'app/(dashboard)/dashboard/super/schools/*', 'app/(dashboard)/dashboard/super/onboarding/page.tsx', 'components/demo/SchoolActionCenter.tsx'	Core multi-tenant admin module
Roles & Access	Create admin/lecturer users, assign school ownership, review status	'super_admin'	'app/(dashboard)/dashboard/super/roles/page.tsx'	Needs RBAC + invitation flow
Subscriptions & Billing	Plan management, fee structure, billing transaction visibility	'super_admin'	'app/(dashboard)/dashboard/super/fees/page.tsx'	Implies payment/billing backend
Faculties & Levels	Faculty health, capacity, level summaries	'school_admin'	'app/(dashboard)/dashboard/school/faculties/page.tsx', 'app/(dashboard)/dashboard/school/levels/page.tsx', 'components/dashboard/FacultyTable.tsx'	Model naming is inconsistent with "schools"
Courses	Platform/school course registry, lecturer assigned courses, class rep mapping	'super_admin', 'school_admin', 'lecturer'	'app/(dashboard)/dashboard/super/courses/page.tsx', 'app/(dashboard)/dashboard/school/courses/page.tsx', 'components/lecturer/AssignedCoursesTable.tsx'	Shared with student/mobile
Lecturer Directory	Lecturer invite, directory, responsibility updates	'super_admin', 'school_admin'	'app/(dashboard)/dashboard/super/lecturers/page.tsx', 'app/(dashboard)/dashboard/school/lecturers/page.tsx', 'components/dashboard/ManagePanel.tsx'	Separate from generic auth users

Students & Admissions	Enrollment oversight, admission queue, student readiness	'super_admin', 'school_admin', 'lecturer'	'app/(dashboard)/dashboard/super/students/page.tsx', 'app/(dashboard)/dashboard/school/students/page.tsx', 'components/demo/AdmissionQueuePanel.tsx'	Shared with student/mobile identity lifecycle
Class Reps & Notifications	Assign class reps, broadcast/notify	'school_admin', 'lecturer'	'components/demo/ClassRepAssignmentBoard.tsx', 'components/lecturer/ClassRepsPanel.tsx', 'components/demo/ScheduleClassForm.tsx'	Shared communication surface
Content Library & Uploads	Platform/school/lecturer content repository, preview/export/upload	all roles with scope limits	'app/(dashboard)/dashboard/*content/page.tsx', 'components/demo/ContentLibraryPanel.tsx', 'components/dashboard/ContentUploadPanel.tsx'	Strong file storage requirement
Assessments	Quiz/assignment/mid-term drafting and publishing	'lecturer'	'app/(dashboard)/dashboard/lecturer/assessments/page.tsx', 'components/demo/AssessmentBuilderPanel.tsx'	Shared with student/mobile submission flow
Class Scheduling	Physical/hybrid/virtual session scheduling	'lecturer'	'app/(dashboard)/dashboard/lecturer/schedule/page.tsx', 'components/demo/ScheduleClassForm.tsx'	Shared with notifications/timetable
Analytics & Reporting	Platform metrics and lecturer performance	'super_admin', 'lecturer'	'app/(dashboard)/dashboard/super/analytics/page.tsx', 'components/lecturer/PerformanceChart.tsx'	Requires aggregation endpoints
Settings / Support / Audit	User preferences, support tickets, system logs	all roles, mostly scoped	'app/(dashboard)/dashboard/lecturer/settings/page.tsx', 'app/(dashboard)/dashboard/lecturer/help/page.tsx', 'components/layout/Sidebar.tsx'	Mostly placeholder but clearly intended

3. Endpoint Requirements By Module

CONVENTIONS USED BELOW

- List endpoints should return '{ data: T[], meta: { page, pageSize, total, totalPages } }'.
- Detail and mutation endpoints should return '{ data: T, meta?: { message: string } }'.
- Dates/times should be ISO 8601.
- All write endpoints should record the acting user and create audit log entries.
- Observed file references show where the UI need is visible. Path names themselves are inferred because the repo contains no live API client.

Authentication / Session

POST

/admin/auth/login

PURPOSE

- Observed: sign an admin user in with email/password and redirect by role.

OBSERVED FROM

- 'components/auth/LoginForm.tsx'
- 'hooks/useAuth.ts'
- 'lib/mock-users.ts'
- 'proxy.ts'

ACCESS

- Public

PAYLOAD

- Observed: 'email'
- Observed: 'password'
- Inferred: 'rememberForDays' or 'rememberMe' from "Keep me authenticated for 30 days"

QUERY PARAMS

- None

RESPONSE

- 'user.id'
- 'user.name'
- 'user.email'
- 'user.role'
- 'user.title'
- 'user.school.id'
- 'user.school.name'
- 'user.avatarUrl'
- 'permissions[]'
- 'accessToken'
- 'refreshToken'
- 'expiresAt'

BACKEND VALIDATION

- valid email format
- password check

- account is active
- role is one of 'super_admin', 'school_admin', 'lecturer'
- related school is not deactivated

SIDE EFFECTS

- create session
- issue refresh token
- audit login event

NOTES

- Observed: frontend expects immediate role-based redirect ('hooks/useAuth.ts').

GET**/admin/auth/me**

PURPOSE

- Observed: hydrate sidebar/topbar/dashboard shell with the authenticated user and school context.

OBSERVED FROM

- 'components/layout/DashboardShell.tsx'
- 'components/layout/Sidebar.tsx'
- 'components/layout/Topbar.tsx'
- 'hooks/useAuth.ts'

ACCESS

- Authenticated 'super_admin', 'school_admin', 'lecturer'

PAYLOAD

- None

QUERY PARAMS

- None

RESPONSE

- 'id', 'name', 'email', 'role', 'title'
- 'school.id', 'school.name'
- 'avatarUrl'
- 'permissions[]'
- 'featureFlags[]'

BACKEND VALIDATION

- valid session or token

SIDE EFFECTS

- optional session touch / last-seen update

NOTES

- Observed: the shell needs both role and school/title labels, not just a bare user id.

POST

/admin/auth/logout

PURPOSE

- Observed: terminate current session from sidebar/topbar logout actions.

OBSERVED FROM

- 'hooks/useAuth.ts'
- 'components/layout/Sidebar.tsx'
- 'components/layout/Topbar.tsx'
- 'components/layout/MobileSidebar.tsx'

ACCESS

- Authenticated users

PAYLOAD

- Inferred: optional 'refreshToken' or 'sessionId'

QUERY PARAMS

- None

RESPONSE

- 'success'

BACKEND VALIDATION

- session belongs to caller

SIDE EFFECTS

- revoke refresh token/session
- audit logout event

NOTES

- Inferred: refresh token revocation is needed even though the demo frontend only clears cookies/local storage.

Platform Summary and Global Search

GET

/admin/platform/summary

PURPOSE

- Observed: feed super-admin overview cards and high-level platform analytics.

OBSERVED FROM

- 'app/(dashboard)/dashboard/super/page.tsx'
- 'app/(dashboard)/dashboard/super/analytics/page.tsx'
- 'app/(dashboard)/dashboard/super/schools/page.tsx'

ACCESS

- 'super_admin'

PAYLOAD

- None

QUERY PARAMS

- Inferred: 'dateFrom', 'dateTo'

RESPONSE

- 'totalSchools'
- 'newSchoolsThisMonth'
- 'totalLecturers'
- 'lecturerVerificationRate'
- 'activeStudents'
- 'studentGrowthRate'
- 'premiumTierCount'
- 'premiumAdoptionRate'
- 'pendingApprovalCount'
- 'attentionNeededCount'

BACKEND VALIDATION

- date range sanity

SIDE EFFECTS

- none

NOTES

- Observed: multiple pages display overlapping summary metrics, so this should be a reusable aggregate endpoint.

GET`/admin/search`**PURPOSE**

- Observed: support global search bars and directory search fields.

OBSERVED FROM

- 'components/layout/Topbar.tsx'
- 'app/(dashboard)/dashboard/super/schools/page.tsx'

ACCESS

- role-scoped by caller

PAYLOAD

- None

QUERY PARAMS

- Observed: 'q'
- Inferred: 'types[]=school|lecturer|student|course|content'
- Inferred: 'limit'

RESPONSE

- grouped results by type with minimal display cards:
 - 'schools[]'
 - 'lecturers[]'
 - 'students[]'
 - 'courses[]'
 - 'content[]'

BACKEND VALIDATION

- minimum query length
- role-based visibility

SIDE EFFECTS

- optional search analytics

NOTES

- Observed: the UI has search affordances across all roles, but no wired state yet.

Schools / Institutions

GET`/admin/schools`

PURPOSE

- Observed: drive the super-admin school directory and school-overview tables.

OBSERVED FROM

- 'app/(dashboard)/dashboard/super/page.tsx'
- 'app/(dashboard)/dashboard/super/schools/page.tsx'
- 'components/demo/SchoolActionCenter.tsx'
- 'lib/mock-data/platform.ts'

ACCESS

- 'super_admin'

PAYLOAD

- None

QUERY PARAMS

- Observed: 'q'
- Observed: 'plan'
- Observed: 'status'
- Inferred: 'page', 'pageSize', 'sort'

RESPONSE

- per school:
 - 'id'
 - 'referenceCode'
 - 'name'
 - 'institutionType'
 - 'location'
 - 'primaryAdminName'
 - 'primaryAdminEmail'
 - 'students'
 - 'lecturers'
 - 'courses'
 - 'capacity'

- 'plan'
- 'status'
- 'deanName'
- meta totals matching cards

BACKEND VALIDATION

- filter enum validation
- pagination bounds

SIDE EFFECTS

- none

NOTES

- Observed: the directory needs both table rows and card totals.

POST`/admin/schools`

PURPOSE

- Observed: onboard a new institution or save a draft.

OBSERVED FROM

- 'app/(dashboard)/dashboard/super/onboarding/page.tsx'
- 'components/demo/SchoolOnboardingForm.tsx'

ACCESS

- 'super_admin'

PAYLOAD

- Observed: 'name'
- Observed: 'contactEmail'
- Observed: 'phone'
- Observed: 'address'
- Observed: 'primaryAdminName'
- Observed: 'primaryAdminEmail'
- Observed: 'capacity'
- Observed: 'status'
- Observed: 'subscriptionPlan'
- Inferred: 'institutionType'
- Inferred: 'referenceCode'

- Inferred: 'billingSetupFee'
- Inferred: 'mode' = 'draft' or 'complete'

QUERY PARAMS

- None

RESPONSE

- created school summary
- onboarding status
- created primary admin user/invitation summary
- subscription assignment summary

BACKEND VALIDATION

- unique school name or reference code
- valid primary admin email
- plan exists
- required fields vary by 'mode'

SIDE EFFECTS

- create tenant/school record
- create draft or active subscription
- create/send primary admin invite
- audit onboarding event

NOTES

- Observed: the UI supports both "Complete Onboarding" and "Save as Draft".

GET

/admin/schools/:schoolId

PURPOSE

- Observed: render the school profile page and modal summary.

OBSERVED FROM

- 'app/(dashboard)/dashboard/super/schools/[id]/page.tsx'
- 'components/demo/SchoolActionCenter.tsx'

ACCESS

- 'super_admin'

PAYLOAD

- None

QUERY PARAMS

- Inferred: 'include=stats,ownership,subscription'

RESPONSE

- 'id', 'name', 'location', 'status'
- 'deanName'
- 'students', 'lecturers', 'courses'
- 'campusMode'
- 'primaryAdmin'
- 'subscription'

BACKEND VALIDATION

- school exists
- caller has access

SIDE EFFECTS

- optional profile-view audit log

NOTES

- Observed: the profile view is more than a flat record; it needs stats and administrative ownership.

PATCH

/admin/schools/:schoolId

PURPOSE

- Inferred: support manage/edit flows from the directory and profile surface.

OBSERVED FROM

- 'app/(dashboard)/dashboard/super/schools/page.tsx'
- 'app/(dashboard)/dashboard/super/schools/[id]/page.tsx'
- 'components/demo/SchoolActionCenter.tsx'

ACCESS

- 'super_admin'

PAYLOAD

- Inferred: editable subset of school profile and plan assignment fields

QUERY PARAMS

- None

RESPONSE

- updated school detail

BACKEND VALIDATION

- immutable/unique fields protected
- deactivated schools may need stricter edits

SIDE EFFECTS

- audit update event
- cache invalidation for school lists/profile

NOTES

- Inferred: "Manage" and "Open Full Profile" imply eventual edit capability even though no edit form is wired yet.

POST`/admin/schools/:schoolId/deactivation`**PURPOSE**

- Observed: deactivate a school with a reason.

OBSERVED FROM

- 'components/demo/SchoolActionCenter.tsx'

ACCESS

- 'super_admin'

PAYLOAD

- Observed: 'reason'
- Inferred: 'effectiveAt'
- Inferred: 'notifyUsers' default true

QUERY PARAMS

- None

RESPONSE

- school status transition summary

BACKEND VALIDATION

- school exists
- caller can deactivate
- reason required

- prevent duplicate active deactivation requests

SIDE EFFECTS

- disable or restrict related admin/lecturer/student access
- notify primary admin
- audit log
- invalidate active sessions if policy requires

NOTES

- Observed: the modal explicitly says deactivation affects admins, lecturers, and students attached to the school.

GET

/admin/schools/:schoolId/export

PURPOSE

- Observed: export a school profile.

OBSERVED FROM

- 'components/demo/SchoolActionCenter.tsx'

ACCESS

- 'super_admin'

PAYLOAD

- None

QUERY PARAMS

- Inferred: 'format=pdf|csv|json'

RESPONSE

- file stream or signed download URL

BACKEND VALIDATION

- school exists
- export format allowed

SIDE EFFECTS

- audit export event
- optional queued generation job

NOTES

- Observed: UI says "Export Profile"; backend should support async generation for large exports.

Roles / Access Management

GET`/admin/users`

PURPOSE

- Observed: list role assignments with filters by role and school.

OBSERVED FROM

- `'app/(dashboard)/dashboard/super/roles/page.tsx'`
- `'lib/mock-data/platform.ts'`

ACCESS

- `'super_admin'`

PAYLOAD

- None

QUERY PARAMS

- Observed: `'q'`
- Observed: `'role'`
- Observed: `'schoolId'`
- Inferred: `'status'`
- Inferred: `'page'`, `'pageSize'`

RESPONSE

- per user:
 - `'id'`
 - `'name'`
 - `'email'`
 - `'phone'`
 - `'staffId'`
 - `'role'`
 - `'school.id'`
 - `'school.name'`
 - `'status'`
 - `'invitedAt'`

- 'approvedAt'

BACKEND VALIDATION

- filter enums
- scoped visibility

SIDE EFFECTS

- none

NOTES

- Observed: list shows both school admins and lecturers in one directory.

POST

/admin/users

PURPOSE

- Observed: create an admin or lecturer account and bind it to a school.

OBSERVED FROM

- 'app/(dashboard)/dashboard/super/roles/page.tsx'

ACCESS

- 'super_admin'

PAYLOAD

- Observed: 'name'
- Observed: 'email'
- Observed: 'phone'
- Observed: 'staffId'
- Observed: 'role'
- Observed: 'schoolId'
- Inferred: 'sendInvite'

QUERY PARAMS

- None

RESPONSE

- created user record
- invite status

BACKEND VALIDATION

- unique email

- unique staff id within system or school, depending policy
- role allowed
- school exists

SIDE EFFECTS

- create auth principal
- send invite/credential email
- audit provisioning event

NOTES

- Observed: copy says "Authorize & Create Account", so approval state and invitation flow should exist.

PATCH

/admin/users/:userId/access

PURPOSE

- Observed/Inferred: update role, school mapping, or approval state.

OBSERVED FROM

- 'app/(dashboard)/dashboard/super/roles/page.tsx'
- 'app/(dashboard)/dashboard/super/lecturers/page.tsx'

ACCESS

- 'super_admin'

PAYLOAD

- Inferred: 'role'
- Inferred: 'schoolId'
- Inferred: 'status'
- Inferred: 'permissions[]'

QUERY PARAMS

- None

RESPONSE

- updated user/access record

BACKEND VALIDATION

- cannot remove last required super admin
- role/school transition rules enforced

SIDE EFFECTS

- session invalidation if access scope changes
- audit log
- notifications to affected user

NOTES

- Observed: "Update Role" and pending/approved states imply post-creation access changes.

Lecturer Directory / Academic Staff

GET`/admin/lecturers`

PURPOSE

- Observed: list lecturers for super-admin oversight and school-admin management.

OBSERVED FROM

- 'app/(dashboard)/dashboard/super/lecturers/page.tsx'
- 'app/(dashboard)/dashboard/school/lecturers/page.tsx'
- 'components/dashboard/ManagePanel.tsx'
- 'components/dashboard/LecturerList.tsx'

ACCESS

- 'super_admin'
- 'school_admin' scoped to own school

PAYLOAD

- None

QUERY PARAMS

- Inferred: 'schoolId'
- Inferred: 'department'
- Inferred: 'verified'
- Inferred: 'page', 'pageSize'

RESPONSE

- 'id', 'name', 'email'
- 'department'
- 'title'
- 'currentResponsibility'
- 'school'

- 'assignedCourseCount'
- 'verifiedStatus'

BACKEND VALIDATION

- role-based scope

SIDE EFFECTS

- none

NOTES

- Observed: super admin needs higher-level responsibility labels; school admin just needs directory + add flow.

POST

/admin/lecturers

PURPOSE

- Observed/Inferred: add a new lecturer from school management or invite workflow.

OBSERVED FROM

- 'app/(dashboard)/dashboard/school/lecturers/page.tsx'
- 'components/dashboard/ManagePanel.tsx'
- 'app/(dashboard)/dashboard/super/lecturers/page.tsx'

ACCESS

- 'super_admin'
- 'school_admin'

PAYLOAD

- Inferred: 'name'
- Inferred: 'email'
- Inferred: 'phone'
- Inferred: 'staffId'
- Inferred: 'department'
- Inferred: 'title'
- Inferred: 'schoolId'
- Inferred: 'sendInvite'

QUERY PARAMS

- None

RESPONSE

- created lecturer record

BACKEND VALIDATION

- required identity and school mapping
- unique email/staff id

SIDE EFFECTS

- create lecturer auth account if missing
- send invite
- audit event

NOTES

- Observed: UI repeatedly exposes "Add New Lecturer" or "Invite Lecturer".

PATCH

/admin/lecturers/:lecturerId

PURPOSE

- Observed/Inferred: update lecturer role/responsibility and review resources.

OBSERVED FROM

- 'app/(dashboard)/dashboard/super/lecturers/page.tsx'

ACCESS

- 'super_admin'
- 'school_admin' for same-school lecturers

PAYLOAD

- Inferred: 'department'
- Inferred: 'title'
- Inferred: 'assignedCourseIds[]'
- Inferred: 'verifiedStatus'

QUERY PARAMS

- None

RESPONSE

- updated lecturer profile

BACKEND VALIDATION

- course assignments belong to same school unless super-admin override

SIDE EFFECTS

- update course ownership
- notify lecturer
- audit reassignment

NOTES

- Observed: "Current Responsibility" is visible in cards, so backend needs a stored responsibility/title field.

Subscriptions / Fees / Billing

GET

/admin/subscriptions/summary

PURPOSE

- Observed: populate subscription/fees top stats.

OBSERVED FROM

- 'app/(dashboard)/dashboard/super/fees/page.tsx'

ACCESS

- 'super_admin'

PAYLOAD

- None

QUERY PARAMS

- Inferred: 'dateFrom', 'dateTo'

RESPONSE

- 'totalAnnualRevenue'
- 'revenueGrowthRate'
- 'activeSubscriptions'
- 'activeRegions'
- 'churnRate'
- 'benchmark'

BACKEND VALIDATION

- date range sanity

SIDE EFFECTS

- none

NOTES

- Observed: the page mixes commercial KPIs with plan management.

GET`/admin/subscriptions/plans`**PURPOSE**

- Observed: list fee tiers and entitlements.

OBSERVED FROM

- 'app/(dashboard)/dashboard/super/fees/page.tsx'
- 'lib/mock-data/platform.ts'

ACCESS

- 'super_admin'

PAYLOAD

- None

QUERY PARAMS

- Inferred: 'status', 'page', 'pageSize'

RESPONSE

- per plan:
 - 'id'
 - 'name'
 - 'price'
 - 'billingInterval'
 - 'features[]'
 - 'isMostPopular'
 - 'status'

BACKEND VALIDATION

- none

SIDE EFFECTS

- none

NOTES

- Observed: entitlements are feature-based, not just price-based.

POST`/admin/subscriptions/plans`

PURPOSE

- Observed/Inferred: create a new institutional plan.

OBSERVED FROM

- 'app/(dashboard)/dashboard/super/fees/page.tsx'

ACCESS

- 'super_admin'

PAYLOAD

- Inferred: 'name'
- Inferred: 'price'
- Inferred: 'billingInterval'
- Inferred: 'features[]'
- Inferred: 'status'

QUERY PARAMS

- None

RESPONSE

- created plan record

BACKEND VALIDATION

- unique plan name
- non-negative price

SIDE EFFECTS

- audit pricing change

NOTES

- Observed: "Create New Plan" exists, but there is no modal yet.

PATCH

/admin/subscriptions/plans/:planId

PURPOSE

- Observed: edit existing plan settings.

OBSERVED FROM

- 'app/(dashboard)/dashboard/super/fees/page.tsx'

ACCESS

- 'super_admin'

PAYLOAD

- Inferred: editable pricing/feature/status fields

QUERY PARAMS

- None

RESPONSE

- updated plan record

BACKEND VALIDATION

- protect plans already attached to active schools with migration rules

SIDE EFFECTS

- audit pricing change
- may queue downstream school entitlement recalculation

NOTES

- Observed: each tier card has an "Edit Plan" action.

GET`/admin/subscriptions/transactions`**PURPOSE**

- Observed: show recent billing transactions/reporting.

OBSERVED FROM

- 'app/(dashboard)/dashboard/super/fees/page.tsx'

ACCESS

- 'super_admin'

PAYLOAD

- None

QUERY PARAMS

- Inferred: 'schoolId', 'planId', 'status', 'dateFrom', 'dateTo', 'page'

RESPONSE

- per transaction:
 - 'id'
 - 'schoolName'
 - 'planName'
 - 'renewalDate'

- 'amount'
- 'currency'
- 'status'

BACKEND VALIDATION

- filter enums

SIDE EFFECTS

- none

NOTES

- Inferred: webhook-backed reconciliation is required even though this page is read-only today.

Faculties / Levels

GET

/admin/schools/:schoolId/faculties

PURPOSE

- Observed: render faculty distribution/health tables for school admins.

OBSERVED FROM

- 'app/(dashboard)/dashboard/school/page.tsx'
- 'app/(dashboard)/dashboard/school/faculties/page.tsx'
- 'components/dashboard/FacultyTable.tsx'

ACCESS

- 'school_admin'
- 'super_admin'

PAYLOAD

- None

QUERY PARAMS

- Observed: level filter exists on the dashboard
- Inferred: 'status', 'q', 'page'

RESPONSE

- per faculty:
 - 'id'
 - 'name'

- 'students'
- 'capacityPercent'
- 'status'
- 'deanName'

BACKEND VALIDATION

- caller can access school

SIDE EFFECTS

- none

NOTES

- Observed: the action menu implies edit/deactivate/profile behavior even though it is not wired.

PATCH

/admin/faculties/:facultyId

PURPOSE

- Inferred: support faculty edit actions from the row action menu.

OBSERVED FROM

- 'components/dashboard/FacultyTable.tsx'
- 'components/ui/ActionMenu.tsx'

ACCESS

- 'school_admin' for own school
- 'super_admin'

PAYLOAD

- Inferred: 'name', 'capacityTarget', 'status', 'deanName'

QUERY PARAMS

- None

RESPONSE

- updated faculty record

BACKEND VALIDATION

- faculty belongs to caller scope

SIDE EFFECTS

- audit update

NOTES

- Observed: action menu exposes 'Edit'.

POST

/admin/faculties/:facultyId/deactivate

PURPOSE

- Inferred: support faculty-level deactivation from row action menu.

OBSERVED FROM

- 'components/ui/ActionMenu.tsx'

ACCESS

- 'school_admin'
- 'super_admin'

PAYLOAD

- Inferred: 'reason'

QUERY PARAMS

- None

RESPONSE

- updated faculty status

BACKEND VALIDATION

- reason required

SIDE EFFECTS

- restrict new course/admission actions under that faculty
- audit log

NOTES

- Observed: action menu exposes 'Deactivate'.

GET

/admin/schools/:schoolId/levels

PURPOSE

- Observed: render academic level summaries and occupancy metrics.

OBSERVED FROM

- 'app/(dashboard)/dashboard/school/levels/page.tsx'

- 'app/(dashboard)/dashboard/super/levels/page.tsx'

ACCESS

- 'school_admin' for scoped school
- 'super_admin'

PAYLOAD

- None

QUERY PARAMS

- Inferred: 'schoolId' optional for super-admin aggregate view

RESPONSE

- per level:
 - 'id'
 - 'name'
 - 'studentCount'
 - 'occupancyPercent'
 - 'note'

BACKEND VALIDATION

- scope checks

SIDE EFFECTS

- none

NOTES

- Observed: school and super-admin level pages use similar but not identical summary cards.

Courses

GET

/admin/courses

PURPOSE

- Observed: power platform course registry, school course management, and lecturer assigned-course views.

OBSERVED FROM

- 'app/(dashboard)/dashboard/super/courses/page.tsx'
- 'app/(dashboard)/dashboard/school/courses/page.tsx'

- 'components/lecturer/AssignedCoursesTable.tsx'
- 'lib/mock-data/platform.ts'
- 'lib/mock-data/courses.ts'

ACCESS

- 'super_admin'
- 'school_admin'
- 'lecturer' for own assignments

PAYLOAD

- None

QUERY PARAMS

- Inferred: 'scope=platform|school|assigned'
- Inferred: 'schoolId'
- Inferred: 'lecturerId=me'
- Observed: 'level'
- Observed: 'module'
- Inferred: 'q', 'page', 'pageSize', 'sortBy=enrollment|performance'

RESPONSE

- per course:
 - 'id'
 - 'code'
 - 'name'
 - 'level'
 - 'module'
 - 'schoolId'
 - 'lecturerIds[]'
 - 'enrollment'
 - 'enrollmentTrend'
 - 'trendType'
 - 'classRep'
 - 'performanceIndex'

BACKEND VALIDATION

- scope filtering by role
- sort/filter enum validation

SIDE EFFECTS

- none

NOTES

- Observed: the same domain object is viewed differently by super admin, school admin, and lecturer.

POST

/admin/courses

PURPOSE

- Observed/Inferred: create a new course or course draft.

OBSERVED FROM

- 'app/(dashboard)/dashboard/super/courses/page.tsx'
- 'app/(dashboard)/dashboard/school/courses/page.tsx'

ACCESS

- 'super_admin'
- 'school_admin'

PAYLOAD

- Inferred: 'code'
- Inferred: 'name'
- Inferred: 'level'
- Inferred: 'module'
- Inferred: 'schoolId'
- Inferred: 'lecturerIds[]'
- Inferred: 'classRepStudentId'
- Inferred: 'status'

QUERY PARAMS

- None

RESPONSE

- created course record

BACKEND VALIDATION

- unique course code within school or platform rules
- school and lecturer assignments valid

SIDE EFFECTS

- expose course in lecturer dashboard
- expose course to student/mobile catalog
- audit event

NOTES

- Observed: "New Course" and "Plan New Course" are present but not wired.

PATCH

/admin/courses/:courseId

PURPOSE

- Inferred: update delivery, assignment, or performance-linked metadata.

OBSERVED FROM

- 'app/(dashboard)/dashboard/school/courses/page.tsx'
- 'components/lecturer/AssignedCoursesTable.tsx'
- 'components/demo/ClassRepAssignmentBoard.tsx'

ACCESS

- 'super_admin'
- 'school_admin'

PAYLOAD

- Inferred: editable fields from course creation plus 'classRepStudentId'

QUERY PARAMS

- None

RESPONSE

- updated course record

BACKEND VALIDATION

- assignments remain within same school/faculty rules

SIDE EFFECTS

- update lecturer and mobile/student views
- audit log

NOTES

- Inferred: class rep assignment is course-linked, so course and class-rep data must stay consistent.

Students / Admissions

GET`/admin/students`**PURPOSE**

- Observed/Inferred: support student oversight and a future student directory.

OBSERVED FROM

- `'app/(dashboard)/dashboard/super/students/page.tsx'`
- `'app/(dashboard)/dashboard/school/students/page.tsx'`
- `'components/demo/AdmissionQueuePanel.tsx'`

ACCESS

- `'super_admin'`
- `'school_admin'`
- `'lecturer'` for relevant course/student scope if needed

PAYLOAD

- None

QUERY PARAMS

- Inferred: `'schoolId'`, `'program'`, `'level'`, `'status'`, `'q'`, `'page'`

RESPONSE

- per student:
 - `'id'`
 - `'name'`
 - `'program'`
 - `'level'`
 - `'status'`
 - `'schoolId'`
 - `'classRepAssignments[]'`

BACKEND VALIDATION

- role-based data scope

SIDE EFFECTS

- none

NOTES

- Observed: "Open Student Directory" exists but no dedicated page is wired yet.

GET`/admin/admissions`**PURPOSE**

- Observed: list the admission queue with reviewable status.

OBSERVED FROM

- `'app/(dashboard)/dashboard/school/admissions/page.tsx'`
- `'app/(dashboard)/dashboard/school/students/page.tsx'`
- `'app/(dashboard)/dashboard/lecturer/enrollment/page.tsx'`
- `'components/demo/AdmissionQueuePanel.tsx'`

ACCESS

- `'school_admin'`
- `'super_admin'`
- `'lecturer'` read-only if intended for course readiness

PAYLOAD

- None

QUERY PARAMS

- Inferred: `'schoolId'`, `'status'`, `'program'`, `'level'`, `'page'`

RESPONSE

- per admission:
 - `'id'`
 - `'studentName'`
 - `'program'`
 - `'level'`
 - `'status'`
 - `'documentsVerified'`
 - `'feeClearanceStatus'`

BACKEND VALIDATION

- scope checks

SIDE EFFECTS

- none

NOTES

- Observed: statuses include document verification and fee-clearance readiness.

POST`/admin/admissions/:admissionId/review`**PURPOSE**

- Observed/Inferred: move an admission item forward for registrar review or approval.

OBSERVED FROM

- 'components/demo/AdmissionQueuePanel.tsx'

ACCESS

- 'school_admin'
- 'super_admin'

PAYLOAD

- Observed/Inferred: 'action'
- Inferred: 'note'
- Inferred: 'nextStatus'

QUERY PARAMS

- None

RESPONSE

- updated admission status

BACKEND VALIDATION

- legal state transition
- reviewer can act for that school

SIDE EFFECTS

- notify applicant/admin stakeholders
- create or activate student profile on approval
- audit log

NOTES

- Observed: UI copy says "marked for registrar review".

Class Reps / Notifications**GET**`/admin/class-reps`**PURPOSE**

- Observed: list class rep assignments per course.

OBSERVED FROM

- 'components/demo/ClassRepAssignmentBoard.tsx'
- 'components/lecturer/ClassRepsPanel.tsx'
- 'lib/mock-data/platform.ts'
- 'lib/mock-data/students.ts'

ACCESS

- 'school_admin'
- 'lecturer'
- 'super_admin' optionally read-only

PAYLOAD

- None

QUERY PARAMS

- Inferred: 'schoolId', 'courseId', 'engagement', 'page'

RESPONSE

- per assignment:
 - 'assignmentId'
 - 'courseId'
 - 'courseCode'
 - 'repStudentId'
 - 'repName'
 - 'repInitials'
 - 'channel'
 - 'engagementStatus'
 - 'active'

BACKEND VALIDATION

- rep belongs to course/school

SIDE EFFECTS

- none

NOTES

- Observed: communication channel is part of the assignment model, not just a student flag.

PUT`/admin/class-reps/:assignmentId`**PURPOSE**

- Observed/Inferred: assign or change a class representative and activation state.

OBSERVED FROM

- 'components/demo/ClassRepAssignmentBoard.tsx'

ACCESS

- 'school_admin'
- 'lecturer' for owned courses

PAYLOAD

- Inferred: 'repStudentId'
- Inferred: 'channel'
- Inferred: 'active'

QUERY PARAMS

- None

RESPONSE

- updated assignment

BACKEND VALIDATION

- student belongs to course
- caller owns the course or school

SIDE EFFECTS

- notify outgoing/incoming rep if changed
- update mobile/student rep privileges
- audit log

NOTES

- Observed: page subtitle says "Assign, review, and activate".

POST`/admin/notifications`**PURPOSE**

- Observed: send targeted or broadcast notifications to class reps/students/support targets.

OBSERVED FROM

- 'components/lecturer/ClassRepsPanel.tsx'
- 'components/demo/ClassRepAssignmentBoard.tsx'
- 'components/demo/ScheduleClassForm.tsx'
- 'app/(dashboard)/dashboard/lecturer/help/page.tsx'

ACCESS

- 'super_admin'
- 'school_admin'
- 'lecturer' scoped to own audiences

PAYLOAD

- Inferred: 'audienceType'
- Inferred: 'audienceIds[]'
- Inferred: 'channel'
- Inferred: 'title'
- Inferred: 'body'
- Inferred: 'relatedCourseId'
- Inferred: 'relatedSessionId'

QUERY PARAMS

- None

RESPONSE

- message dispatch summary with recipient counts

BACKEND VALIDATION

- audience exists and caller may target them
- message size/channel rules

SIDE EFFECTS

- queue email/push/in-app/Telegram delivery
- analytics/audit log

NOTES

- Observed: the UI mentions Telegram broadcast, email list, class portal, and class-session audience options.

Content Library / Uploads

GET**/admin/content****PURPOSE**

- Observed: list platform, school, or lecturer content items with preview/export actions.

OBSERVED FROM

- 'app/(dashboard)/dashboard/super/content/page.tsx'
- 'app/(dashboard)/dashboard/school/content/page.tsx'
- 'app/(dashboard)/dashboard/lecturer/content/page.tsx'
- 'components/demo/ContentLibraryPanel.tsx'

ACCESS

- all roles with scope restrictions

PAYLOAD

- None

QUERY PARAMS

- Inferred: 'scope=platform|school|course|lecturer'
- Observed: 'visibility'
- Observed: 'owner'
- Inferred: 'q', 'updatedSince', 'page'

RESPONSE

- per content item:
 - 'id'
 - 'title'
 - 'ownerLabel'
 - 'visibility'
 - 'updatedAt'
 - 'fileType'
 - 'previewUrl'
 - 'downloadUrl'
 - 'schoolId'
 - 'courseId'

BACKEND VALIDATION

- visibility scoped to caller

SIDE EFFECTS

- none

NOTES

- Observed: visibility values include platform, school, and faculty-like scope labels.

POST

/admin/content

PURPOSE

- Observed/Inferred: upload a new resource to the repository.

OBSERVED FROM

- 'components/demo/ContentLibraryPanel.tsx'
- 'components/dashboard/ContentUploadPanel.tsx'

ACCESS

- 'super_admin'
- 'school_admin'
- 'lecturer'

PAYLOAD

- Inferred: 'title'
- Inferred: 'visibility'
- Inferred: 'ownerLabel'
- Inferred: 'schoolId'
- Inferred: 'courseId'
- Inferred: 'files[]' or upload-token handshake

QUERY PARAMS

- None

RESPONSE

- created content metadata
- upload processing status

BACKEND VALIDATION

- file type/size rules
- caller scope matches chosen visibility

SIDE EFFECTS

- store file
- virus scan
- preview generation
- audit log
- optional mobile/content index sync

NOTES

- Observed: the UI has upload affordances but no metadata form yet, so the frontend will need expansion.

GET

/admin/content/:contentId/preview

PURPOSE

- Observed: preview a content item.

OBSERVED FROM

- 'components/demo/ContentLibraryPanel.tsx'

ACCESS

- role-scoped

PAYLOAD

- None

QUERY PARAMS

- None

RESPONSE

- preview stream or signed preview URL

BACKEND VALIDATION

- access to this content item

SIDE EFFECTS

- audit preview event

NOTES

- Inferred: generated previews or proxy streaming will likely be needed for PDFs/docs.

GET

/admin/content/:contentId/download

PURPOSE

- Observed: export/download a content item.

OBSERVED FROM

- 'components/demo/ContentLibraryPanel.tsx'

ACCESS

- role-scoped

PAYLOAD

- None

QUERY PARAMS

- None

RESPONSE

- file stream or signed download URL

BACKEND VALIDATION

- access to this content item

SIDE EFFECTS

- audit download event

NOTES

- Observed: the UI labels this action as 'Export'.

Assessments

GET

/admin/assessments

PURPOSE

- Inferred: list existing lecturer assessments and drafts.

OBSERVED FROM

- 'app/(dashboard)/dashboard/lecturer/assessments/page.tsx'
- 'components/demo/AssessmentBuilderPanel.tsx'

ACCESS

- 'lecturer'
- 'school_admin' optionally for oversight

PAYLOAD

- None

QUERY PARAMS

- Inferred: 'courseId', 'status', 'type', 'lecturerId=me'

RESPONSE

- per assessment:
 - 'id'
 - 'courseId'
 - 'courseCode'
 - 'type'
 - 'submissionMode'
 - 'durationMinutes'
 - 'questionCount'
 - 'autoGradingEnabled'
 - 'status'
 - 'publishAt'

BACKEND VALIDATION

- lecturer ownership/scope

SIDE EFFECTS

- none

NOTES

- Inferred: once real data exists, lecturers will need to revisit drafts and published assessments.

POST

/admin/assessments

PURPOSE

- Observed: publish an assessment draft.

OBSERVED FROM

- 'components/demo/AssessmentBuilderPanel.tsx'

ACCESS

- 'lecturer'

PAYLOAD

- Observed: 'courseCode' or 'courseId'

- Observed: 'assessmentType'
- Observed: 'submissionMode'
- Observed/Inferred: 'questionCount'
- Observed/Inferred: 'durationMinutes'
- Observed/Inferred: 'autoGradingEnabled'
- Inferred: 'status=draft|published'

QUERY PARAMS

- None

RESPONSE

- created assessment

BACKEND VALIDATION

- course belongs to lecturer
- allowed type/mode combinations
- publish rules for timed CBT

SIDE EFFECTS

- notify students / reps
- create submission window
- queue auto-grading config if enabled
- audit log

NOTES

- Observed: auto-grading, timer, and question-count concepts are product-level features, not presentational only.

Class Scheduling

GET`/admin/class-sessions`

PURPOSE

- Inferred: list scheduled/draft class sessions for lecturer timetable surfaces.

OBSERVED FROM

- 'app/(dashboard)/dashboard/lecturer/schedule/page.tsx'
- 'components/demo/ScheduleClassForm.tsx'

ACCESS

- 'lecturer'
- 'school_admin' optionally scoped

PAYLOAD

- None

QUERY PARAMS

- Inferred: 'lecturerId=me', 'courseId', 'dateFrom', 'dateTo', 'status'

RESPONSE

- per session:
 - 'id'
 - 'courseId'
 - 'courseCode'
 - 'startsAt'
 - 'venueOrLink'
 - 'deliveryMode'
 - 'notifyAudience'
 - 'status'

BACKEND VALIDATION

- scope checks

SIDE EFFECTS

- none

NOTES

- Inferred: drafts are needed because the UI has "Save Draft".

POST

/admin/class-sessions

PURPOSE

- Observed: schedule a physical, hybrid, or virtual class and notify participants.

OBSERVED FROM

- 'components/demo/ScheduleClassForm.tsx'

ACCESS

- 'lecturer'

PAYLOAD

- Observed: 'courseCode' or 'courseId'
- Observed: 'venueOrLink'
- Observed: 'date'
- Observed: 'time'
- Observed: 'deliveryMode'
- Observed: 'notifyAudience'
- Inferred: 'status=draft|scheduled'

QUERY PARAMS

- None

RESPONSE

- created session record

BACKEND VALIDATION

- course belongs to lecturer
- future datetime
- delivery mode compatible with venue/link data

SIDE EFFECTS

- create timetable item
- queue notifications
- audit log

NOTES

- Observed: supported audience options are 'Entire Class', 'Class Reps Only', and 'Registered Students + Reps'.

Analytics / Performance

GET`/admin/analytics/platform`

PURPOSE

- Observed: power the dedicated super-admin analytics page and school attention summaries.

OBSERVED FROM

- 'app/(dashboard)/dashboard/super/analytics/page.tsx'
- 'app/(dashboard)/dashboard/super/page.tsx'

ACCESS

- 'super_admin'

PAYLOAD

- None

QUERY PARAMS

- Inferred: 'dateFrom', 'dateTo', 'schoolId'

RESPONSE

- high-level KPIs plus:
 - 'schoolsLive'
 - 'studentsActive'
 - 'premiumAdoptionRate'
 - 'attentionNeededCount'
 - 'networkHealthSummary'

BACKEND VALIDATION

- date range sanity

SIDE EFFECTS

- none

NOTES

- Observed: this is narrative reporting, not only raw counts.

GET

/admin/analytics/courses/performance

PURPOSE

- Observed: feed lecturer performance charts and metrics.

OBSERVED FROM

- 'app/(dashboard)/dashboard/lecturer/page.tsx'
- 'app/(dashboard)/dashboard/lecturer/performance/page.tsx'
- 'components/lecturer/PerformanceChart.tsx'

ACCESS

- 'lecturer'
- 'school_admin' optionally scoped

PAYLOAD

- None

QUERY PARAMS

- Observed/Inferred: 'period=last_30_days'
- Inferred: 'courseId'
- Inferred: 'scope=me|school'

RESPONSE

- 'series[]' for chart
- 'avgAttendance'
- 'pendingReviews'
- 'facultyRank'

BACKEND VALIDATION

- scope ownership

SIDE EFFECTS

- none

NOTES

- Observed: the dashboard uses both chart data and summary KPIs.

Support / Settings / Audit Logs

POST

/admin/support/tickets

PURPOSE

- Observed/Inferred: open a support ticket or escalation request.

OBSERVED FROM

- 'app/(dashboard)/dashboard/lecturer/help/page.tsx'
- 'components/layout/Sidebar.tsx'
- 'components/layout/MobileSidebar.tsx'

ACCESS

- all authenticated roles

PAYLOAD

- Inferred: 'subject'
- Inferred: 'category'
- Inferred: 'description'
- Inferred: 'severity'

- Inferred: 'relatedCourseId'

QUERY PARAMS

- None

RESPONSE

- created ticket summary

BACKEND VALIDATION

- required description/category

SIDE EFFECTS

- notify support queue
- audit ticket creation

NOTES

- Observed: help/support exists in both lecturer and super-admin navigation, but no real ticket UI is wired yet.

GET

/admin/me/settings

PURPOSE

- Observed: load lecturer settings categories and saved preferences.

OBSERVED FROM

- 'app/(dashboard)/dashboard/lecturer/settings/page.tsx'

ACCESS

- authenticated user

PAYLOAD

- None

QUERY PARAMS

- None

RESPONSE

- 'notificationPrefs'
- 'securityPrefs'
- 'workspacePrefs'

BACKEND VALIDATION

- caller identity

SIDE EFFECTS

- none

NOTES

- Observed: settings categories are Notifications, Security, and Workspace Preferences.

PATCH`/admin/me/settings`**PURPOSE**

- Inferred: persist user settings once the page becomes interactive.

OBSERVED FROM

- 'app/(dashboard)/dashboard/lecturer/settings/page.tsx'

ACCESS

- authenticated user

PAYLOAD

- Inferred: partial settings object

QUERY PARAMS

- None

RESPONSE

- updated settings

BACKEND VALIDATION

- schema validation per settings section

SIDE EFFECTS

- audit preference change if security-sensitive

NOTES

- Inferred: security settings may need re-auth or step-up auth.

GET`/admin/audit-logs`**PURPOSE**

- Observed/Inferred: provide system logs and administrative action history.

OBSERVED FROM

- 'components/layout/Sidebar.tsx'

- 'components/layout/MobileSidebar.tsx'
- 'components/demo/SchoolActionCenter.tsx'

ACCESS

- 'super_admin'
- potentially 'school_admin' with scoped logs

PAYLOAD

- None

QUERY PARAMS

- Inferred: 'actorId', 'module', 'action', 'dateFrom', 'dateTo', 'page'

RESPONSE

- per log:
 - 'id'
 - 'actor'
 - 'action'
 - 'module'
 - 'targetType'
 - 'targetId'
 - 'timestamp'
 - 'metadata'

BACKEND VALIDATION

- scope checks

SIDE EFFECTS

- none

NOTES

- Observed: "System Logs" exists in navigation but is currently a toast-only placeholder.

4. Admin-Only Workflows

- Observed: institution onboarding with primary-admin creation and plan assignment ('app/(dashboard)/dashboard/super/onboarding/page.tsx', 'components/demo/SchoolOnboardingForm.tsx').
- Observed: school directory management, profile inspection, export, and deactivation ('app/(dashboard)/dashboard/super/schools/page.tsx', 'components/demo/SchoolActionCenter.tsx').

- Observed: super-admin user provisioning and school ownership mapping for school admins and lecturers ('app/(dashboard)/dashboard/super/roles/page.tsx').
- Observed: subscription tier creation/editing and transaction review ('app/(dashboard)/dashboard/super/fees/page.tsx').
- Observed/Inferred: faculty status management and capacity balancing ('components/dashboard/FacultyTable.tsx', 'components/ui/ActionMenu.tsx').
- Observed/Inferred: course planning/creation and course-to-class-rep mapping ('app/(dashboard)/dashboard/school/courses/page.tsx', 'components/demo/ClassRepAssignmentBoard.tsx').
- Observed/Inferred: lecturer invite/reassignment and responsibility updates ('app/(dashboard)/dashboard/super/lecturers/page.tsx', 'app/(dashboard)/dashboard/school/lecturers/page.tsx').
- Observed: admission queue review and registrar escalation ('components/demo/AdmissionQueuePanel.tsx').
- Observed: class rep activation and targeted broadcast notifications ('components/demo/ClassRepAssignmentBoard.tsx', 'components/lecturer/ClassRepsPanel.tsx').
- Observed: content preview, export, and upload ('components/demo/ContentLibraryPanel.tsx', 'components/dashboard/ContentUploadPanel.tsx').
- Observed: assessment draft publishing ('components/demo/AssessmentBuilderPanel.tsx').
- Observed: class scheduling with audience notification options ('components/demo/ScheduleClassForm.tsx').
- Observed/Inferred: support ticket submission and audit-log review ('app/(dashboard)/dashboard/lecturer/help/page.tsx', 'components/layout/Sidebar.tsx').

5. Shared Backend With Mobile / Student App

The actual student/mobile repository is not present here, so the items below are inferred from this admin repo plus the existence of 'generate_student_pdf.py' and 'generate_backend_pdf.py'.

Shared Area	What mobile/student likely consumes	What admin app creates/updates	Sync events required
---	---	---	---
Courses	course catalog, assigned lecturer, level/module, performance-ready metadata	course creation, course edits, lecturer assignment, class rep mapping	course created/updated, class rep changed
Student identity/enrollment	approved student account, level/program, active status	admission review, student activation, school-level oversight	admission approved/rejected, school deactivated

Content library	lecture notes, policy docs, orientation packs, downloads/previews	content upload, visibility changes, exportable resources	content uploaded, content updated, content archived
Assessments	published quizzes/assignments, submission windows, auto-grading rules	assessment draft/publish	assessment published, assessment updated
Class reps & notifications	rep privileges, rep-targeted messages, course announcements	rep assignment, broadcasts, schedule notifications	rep assigned/revoked, notification queued/delivered
Class sessions	timetable items, venue/links, session reminders	lecturer scheduling	session scheduled, session updated/cancelled
School/level metadata	school availability, faculty/level structure, access scope	onboarding, faculty status, level configuration	school created, school deactivated, faculty status changed

6. Non-Functional Backend Requirements

- RBAC / permission checks
 - Observed: roles are only 'super_admin', 'school_admin', 'lecturer' ('types/index.ts', 'lib/navigation.ts').
 - Inferred: backend must enforce tenant scope, not just role strings, especially for school-bound lecturers/admins.
- Session management / token refresh / secure auth
 - Observed: demo auth stores full user objects in 'localStorage' and cookies ('hooks/useAuth.ts').
 - Inferred: production backend needs access tokens, refresh tokens, revocation, password hashing, optional device/session tracking, and secure cookie strategy.
- Audit logging
 - Observed: there are destructive/admin-sensitive actions like school deactivation, access changes, plan edits, exports, support tickets ('components/demo/SchoolActionCenter.tsx', 'app/(dashboard)/dashboard/super/roles/page.tsx', 'app/(dashboard)/dashboard/super/fees/page.tsx').
 - Inferred: every write path should emit audit events.
- Rate limiting
 - Inferred: required for login, search, support ticket creation, notification sends, and upload endpoints.
- Caching strategy
 - Inferred: cache platform metrics, school summaries, course lists, and content metadata; aggressively bust caches on mutations.

- Search indexing
 - Observed: global and directory search fields exist ('components/layout/Topbar.tsx', 'app/(dashboard)/dashboard/super/schools/page.tsx').
 - Inferred: backend should index schools, users, students, courses, and content.
- Pagination and filtering standards
 - Observed: school directory has pagination and plan/status filters; role screen has role/school filters.
 - Inferred: use consistent 'page', 'pageSize', 'sort', 'q', and filter enums across modules.
- File upload and storage rules
 - Observed: content upload and drag-drop affordances exist ('components/dashboard/ContentUploadPanel.tsx', 'components/demo/ContentLibraryPanel.tsx').
 - Inferred: enforce MIME/size rules, antivirus scanning, preview generation, signed URLs, retention rules, and private/public visibility support.
- Background jobs / queues
 - Inferred: needed for notification dispatch, export generation, file processing, assessment publishing side effects, and possibly billing reconciliation.
- Email / push / in-app notifications
 - Observed: the UI repeatedly implies sending invites, class notifications, broadcast messages, and support escalations.
 - Inferred: unified notification service with channel-specific delivery logs is required.
- Payment webhook handling
 - Observed: plan tiers and billing transaction lists exist ('app/(dashboard)/dashboard/super/fees/page.tsx').
 - Inferred: backend needs subscription lifecycle state, invoice/renewal reconciliation, and webhook idempotency.
- Monitoring / observability / error tracking
 - Inferred: admin workflows touch auth, billing, uploads, and notifications; backend needs structured logs, tracing, alerts, and per-tenant diagnostics.
- Soft delete / archival / retention
 - Observed: school and faculty "deactivate" flows exist.
 - Inferred: use reversible soft-deactivation and preserve audit/export history.
- Import/export requirements
 - Observed: school profile export and multiple reporting screens exist.
 - Inferred: async export jobs, signed downloads, and at least CSV/PDF support are needed.

- Idempotency for sensitive admin actions
 - Inferred: required for deactivation, account provisioning, plan edits, session scheduling, and assessment publishing to avoid duplicate writes.

7. Missing or Placeholder Frontend Flows

Area	What the frontend currently does	Backend capability eventually needed	Frontend also needs changes later?
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Auth	uses 'MOCK_USERS', local storage, cookies; no real token handling ('hooks/useAuth.ts', 'lib/mock-users.ts')	real login/logout/me/refresh/password reset	Yes
Forgot password	button only shows a demo toast ('components/auth/LoginForm.tsx')	password reset request/confirm flow	Yes
Search and filters	search fields, plan/status filters, pagination buttons are uncontrolled UI only ('components/layout/Topbar.tsx', 'app/(dashboard)/dashboard/super/schools/page.tsx')	searchable/paginated list endpoints	Yes
School onboarding	hardcoded fields, no local state binding, no submit handler ('app/(dashboard)/dashboard/super/onboarding/page.tsx')	school create/draft/invite/subscription APIs	Yes
Duplicate onboarding concept	'components/demo/SchoolOnboardingForm.tsx' exists but is not the real routed form	same as above	Yes
Role creation	static form and static table ('app/(dashboard)/dashboard/super/roles/page.tsx')	user provisioning, approval, filtering	Yes
Fees/plans	create/edit plan buttons are placeholders ('app/(dashboard)/dashboard/super/fees/page.tsx')	subscription CRUD + billing reporting	Yes
Faculty actions	row action menu is generic and unwired ('components/dashboard/FacultyTable.tsx', 'components/ui/ActionMenu.tsx')	faculty edit/deactivate/profile endpoints	Yes

Content upload	upload buttons only show toasts; no file input metadata/progress ('components/dashboard/ContentUploadPanel.tsx', 'components/demo/ContentLibraryPanel.tsx')	upload, preview, download, storage processing	Yes
Admission review	review button is a toast only ('components/demo/AdmissionQueuePanel.tsx')	admission workflow state machine	Minimal UI changes, but still needed
Class rep assignment	notify actions are toast-only; no reassignment form ('components/demo/ClassRepAssignmentBoard.tsx')	rep assignment + notification dispatch	Yes
Assessment builder	static selects and informational cards only ('components/demo/AssessmentBuilderPanel.tsx')	create draft/publish assessment	Yes
Class scheduling	no persisted schedule or list; submit/save are toasts ('components/demo/ScheduleClassForm.tsx')	class session create/list/update	Yes
Support/system logs	support/log buttons only show toasts ('app/(dashboard)/dashboard/lecturer/help/page.tsx', 'components/layout/Sidebar.tsx')	support tickets + audit log APIs	Yes
Settings	read-only category cards ('app/(dashboard)/dashboard/lecturer/settings/page.tsx')	settings GET/PATCH	Yes

8. Suggested Backend Build Order

Phase 1: minimum usable admin backend

- Build real auth/session endpoints and RBAC.
- Build school list/detail/create/deactivate.
- Build user provisioning and lecturer directory endpoints.
- Build read endpoints for courses, faculties, levels, students, and admissions.
- Build audit logging early so every later mutation is traceable.

Phase 2: core academic operations

- Build admission review transitions and student activation.

- Build course create/update plus class rep assignment.
- Build content library list/upload/preview/download.
- Build class session scheduling and generic notification dispatch.
- Build lecturer assessment create/list flow.

Phase 3: commercial/reporting/polish

- Build subscription plan CRUD, transactions, and webhook-backed billing state.
- Build platform analytics aggregates and search indexing.
- Build support tickets, settings persistence, export jobs, and richer audit-log browsing.
- Add background processing, observability, and idempotency hardening.

9. Open Questions / Assumptions

- Observed conflict: the term "school" is overloaded. In mock data it sometimes means institution/campus and sometimes faculty-like units ('lib/mock-data/platform.ts', 'app/(dashboard)/dashboard/school/faculties/page.tsx', 'lib/navigation.ts'). The backend data model must decide whether the hierarchy is 'platform -> institution -> faculty -> department' or something flatter.
- Observed conflict: school-admin nav label 'SCHOOLS' actually routes to '/dashboard/school/faculties' ('lib/navigation.ts').
- Observed conflict: subscription mock pricing strings say '/ month' while the UI copy presents annual billing language ('lib/mock-data/platform.ts', 'app/(dashboard)/dashboard/super/fees/page.tsx').
- Observed conflict: several card totals do not match mock datasets exactly, so they should be treated as presentational placeholders, not authoritative schemas ('app/(dashboard)/dashboard/school/page.tsx', 'lib/mock-data/platform.ts').
- Inferred assumption: lecturer users are both auth principals and academic staff profiles; the backend may need separate 'User' and 'LecturerProfile' models even though the frontend collapses them.
- Inferred assumption: content visibility needs at least platform-level, school-level, and course-level scopes; the current UI labels are not yet normalized.
- Observed: no actual student/mobile repo is present here, so shared-contract notes are inferred from admin screens and root documentation scripts, not cross-repo verified.
- Observed: no README, '.env.example', or service client files were found, so backend URL conventions, upload provider choice, payment provider choice, and notification vendor choice remain open.